

Hyunsik Jeon

Designing End-to-End AI Systems — Portfolio

Portfolio: <https://portfolio-app-5ca9.vercel.app/>

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FEATURED PROJECT 01

IAMX6 — Naver Blog Automation Desktop App Hyunsik Jeon (Solo · Private Repo)

Electron · Puppeteer-core · Node.js · AI/LLM · electron-builder (Windows portable .exe / macOS dmg)

Key takeaway: a Naver blog growth tool shipped as an Electron desktop app (not a web app). Puppeteer drives the real browser, an AI model writes context-aware comments, and mutual-neighbor requests, comment/like automation, visitor auto-reply, and blog analytics all ship as one portable .exe.

Technical Pipeline Detail

- 1. Desktop architecture** Electron main/preload/renderer cleanly separates UI from automation backend.
- 2. Browser automation** Puppeteer-core drives a real Naver session (mutual-neighbor, commenting, likes).
- 3. AI comment generation** AI writes context-aware comments with tone presets (friendly / polite / short).
- 4. Blog analytics** Blog DNA diagnosis, keyword analysis, and post-level analytics modules.
- 5. Distribution** electron-builder packages a single Windows portable .exe plus macOS dmg.

FEATURED PROJECT 02

KOKKOK Garden — K-Beauty Multi-region E-commerce

Hyunsik Jeon (Solo)

Next.js 16 · TypeScript · Supabase · Tailwind CSS 4 · OpenAI GPT-4o mini · Vercel

Live: <https://kokv2.vercel.app>

Key takeaway: stood up a full stack — frontend → backend → DB → chatbot — in days via vibe coding. Korean users are routed to a purchasable `/kr`` store; international users hit a viewing + AI-chatbot `/gl`` store. GPT-4o mini auto-translates product data into 5 languages.

Technical Pipeline Detail

- 1. Multi-region routing** KR/GL split via ``x-user-country`` header, six language routes.
- 2. AI translation** GPT-4o mini translates products into 5 languages (EN/CN/JP/VN/TH), cached 24h via ``unstable_cache``.
- 3. AI chatbot** K-beauty consultation chatbot embedded in the GL store.
- 4. Unified backend** Supabase (PostgreSQL + Auth + Storage) handles admin auth, product CRUD, image upload.
- 5. Deploy** Vercel (<https://kokv2.vercel.app>).

FEATURED PROJECT 03

AIIM — AI Influencer Match

Hyunsik Jeon (Solo)

Python · FastAPI · Next.js 14 · PostgreSQL (Cloud SQL) · Selenium · YouTube Data API · Meta Graph API · LLM · GCP Cloud Run

End-to-end cloud platform ingesting 5,300+ creators across YouTube + Instagram + TikTok in 14 languages, matched to brands via LLM reasoning. Multilingual hashtag discovery surfaced 400 new YouTube creators in one 30-min run and extracted 323 IG business emails. Runs 24/7 on Google Cloud (Cloud Run + Cloud SQL). **Live:** aiim-frontend-559416574965.asia-northeast3.run.app · **GitHub:** github.com/jeonhs9110/AIIM-Showcase

Technical Pipeline Detail

- 1. Multi-platform ingestion** YouTube (1,584) · Instagram (3,735) · TikTok (16) — 5,300+ creators
- 2. Multilingual hashtag discovery** Searches native-language tags (광고 · スポンサー · رعاية ...) to escape English-query convergence
- 3. LLM enrichment** Country, language, brand-collab history, vertical, contacts — with prompt caching for cost control
- 4. Recommendation engine** Plain-English brief → ranked shortlist with rationale · demo-mode gating prevents paid-API burn on the public URL
- 5. Human-shaped Selenium** µs dwell randomisation + 5–15 min AFK scheduler → months of uninterrupted runs
- 6. Infra** GCP asia-northeast3 — Cloud Run + Cloud SQL Postgres + Cloud Build CI

Top Award — IBM & RedHat AI Transformation Program

Vaccine Daily (AI News Pipeline)

Hyunsik Jeon and 2 others

React · FastAPI · PostgreSQL · ChromaDB · GPT-4o-mini · ko-sroberta · HDBSCAN · GraphRAG · Docker

An automated news intelligence pipeline that crawls 10 major media outlets every 5 minutes and uses a multi-agent workflow to generate verified, hallucination-free personalized reports.

Live: <https://jeonhs9110.github.io/vaccine-daily/> · GitHub: github.com/jeonhs9110/vaccine-daily

Technical Pipeline Detail

1. Crawling

Real-time automated data collection from 10 media outlets via Selenium & BS4.

2. Clustering

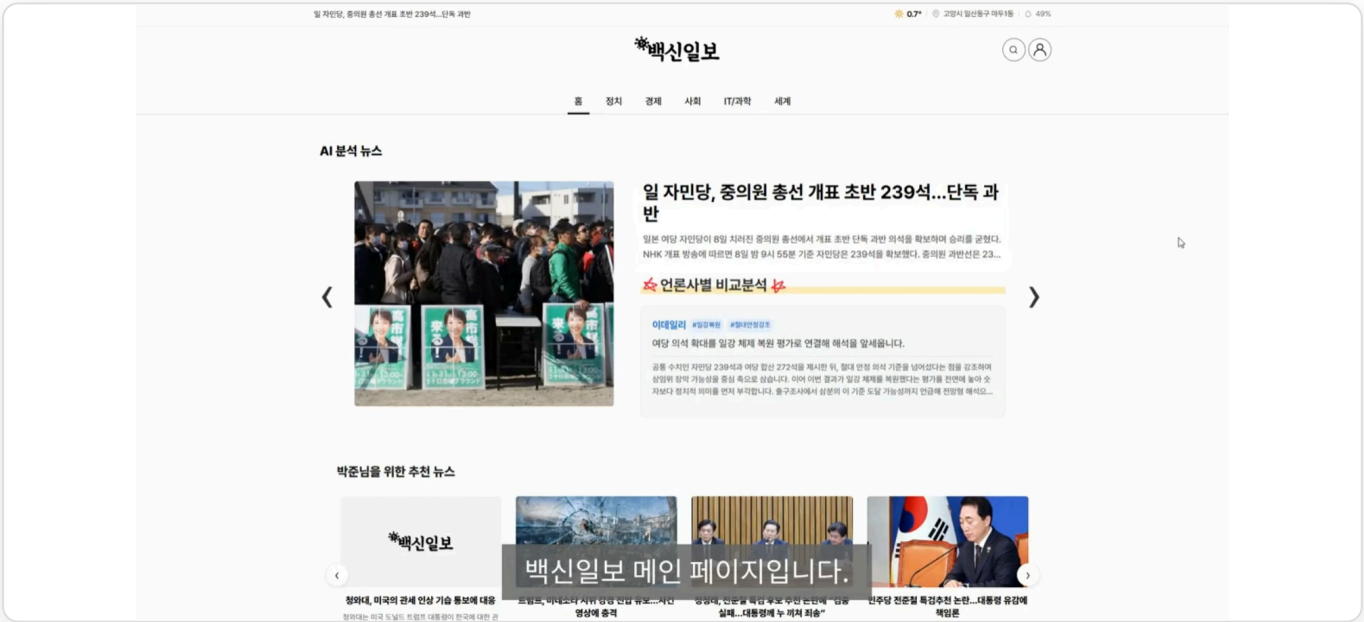
Token-efficient group analysis of similar news using HDBSCAN and ko-sroberta.

3. AI News Article Generator

Multi-Agent AI (Writer → Critic → Editor) for factual verification and summarization.

4. GraphRAG

Entity-Relation extraction and Knowledge Graph construction for Media Tone of Voice analysis.



FEATURED PROJECT 05

FOBO AI — European Football Prediction

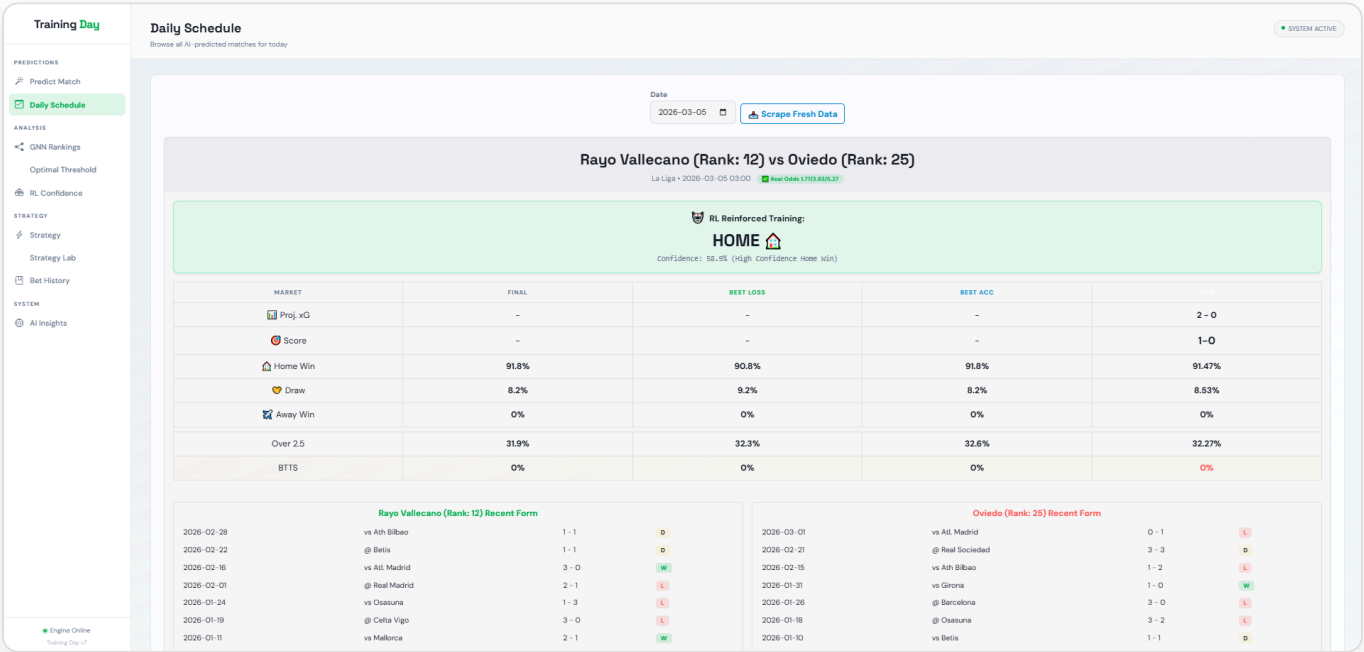
Hyunsik Jeon (Personal Use)

Python · PyTorch · Transformer · Graph Attention Network · PPO RL · Dixon-Coles · XGBoost · LightGBM · Flask · Google Cloud

End-to-end ML system chaining PyTorch Transformer + GAT + Dixon-Coles correction + PPO RL + XGBoost/LightGBM into an 8-stage pipeline. Across 13 European leagues: ~90% hit rate at ensemble $\geq 77\%$ + favourite filter; ~95% when the PPO agent independently agrees at $\geq 90\%$. Deployed on Google Cloud via a 2-VM architecture for ~\$26/month. **Live:** <http://34.64.147.124/> · **GitHub:** github.com/jeonhs9110/epl-showcase

8-Step System Architecture

- 1. Scrape** Parallel Flashscore crawl across 13 leagues (incremental re-runs)
- 2. Validation** Integrity checks catch broken scrapes before they poison training
- 3. Seq optimisation** Per-league look-back window (Premier ~10, Championship ~5)
- 4. DL training** Transformer + GAT + Dixon-Coles NLL — infers defensive posture in weak-vs-strong matchups
- 5. PPO RL** Action $\in \{\text{Home, Draw, Away, Pass}\}$ with Kelly-Criterion-shaped reward
- 6. Hybrid ensemble** XGBoost + LightGBM on DL embeddings
- 7. Calibration** Platt scaling aligns reported probability with empirical hit rate
- 8. Flask API** Models preloaded, <200 ms per request, GPT-4o-mini chatbot scoped to today's matches



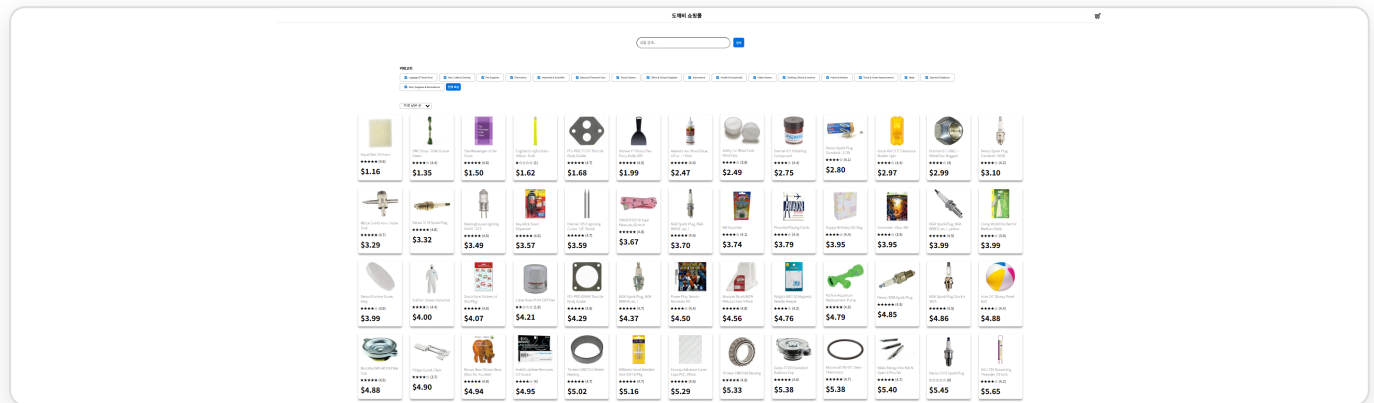
OTHER PROJECTS

Goblin Shopping Mall (E-commerce SPA)

Hyunsik Jeon and 3 others

React · JavaScript · CSS · Vercel

A fully-featured e-commerce platform with 18+ category filters and real-time search. Focused on intuitive UX and high-performance frontend state management.

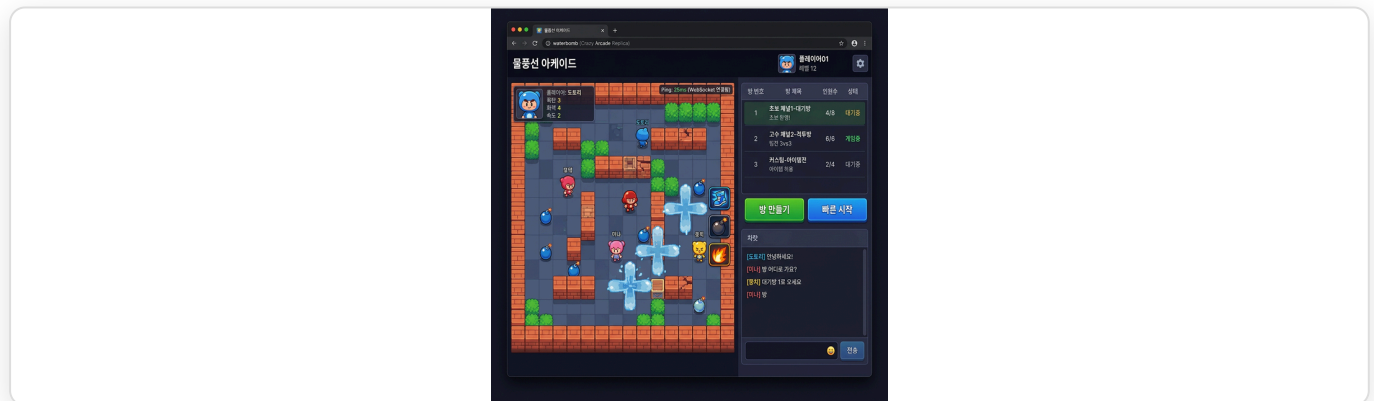


Crazy Arcade Web Replica

Hyunsik Jeon and 3 others

React · FastAPI · WebSocket · MySQL

A real-time multiplayer game synchronization project using WebSockets. Implemented complex physics and room-state management for smooth gameplay.



My Diary (Emotion Log App)

(Personal Use for Education)

React · Context API · localStorage

A personal journal application focusing on frontend state persistence and custom UI interactions using browser's local storage.